CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
CHANGA

M.Tech. EE Sem – I Examination, 2013-14
EE - 701 Engineering Optimization Techniques

Date: 30/12/2013, Monday **Time: 10 a.m. to 1 p.m** **Maximum Marks: 70**

Instructions:

1. Attempt all questions from both sections.
2. Use separate answer book for each sections.
3. Figures to the right indicate **full** marks.
4. Assume suitable additional data, if necessary

SECTION – I

Q-1 A. Use graphical method to solve following Linear Programming Problem [7]
Max. $Z=2x_1+4x_2$
s.t. $x_1+2x_2\leq 5$, $x_1+x_2\leq 4$, $x_1,x_2\geq 0$

B. Solve the following LPP using two-phase method OR Big-M method [7]
Max. $Z=3x_1-x_2$
Subject to: $2x_1+x_2\geq 2$, $x_1+3x_2\leq 3$, $x_2\leq 4$, $x_1, x_2\geq 0$

Q-2 A. Using Steepest Descent Method OR Newton's Method, Minimize following function and Perform only one iteration [7]
Min $f(x)=6x_1^2+3x_2^2+4x_1x_2$

B. Using mathematical equations and graphs, explain difference between Exterior penalty function method and Barrier method [7]

Q-3 Write algorithmic steps of an Interior point method for linear programming. [7]

SECTION – II

Q-4 A. Find out minimum value of the given function using Kuhn-Tucker conditions
Min $f(x_1, x_2) = (x_1-1)^2+(x_2-5)^2$
Subject to: $-x_1^2+x_2 \le 4$, $-(x_1-2)^2+x_2 \le 3$, x_1, x_2 are non-negative [7]

B. Explain: Covariance and Correlation. [4]

C. Define: Positive Semi-definiteness of a matrix [3]

Q-5 A. Using Non-linear programming, minimize following function:
Perform only one iteration
Min $Z=-2x_1-x_2+x_1^2$
Subject to: $-2x_1-3x_2 \ge -6$, $-2x_1-x_2 \ge -4$, $x_1, x_2 \ge 0$ [10]

B. Using Least Square method, solve following equations:
$2x-y=2$, $x+2y=1$, $x+y=4$ [4]

Q-6 A. Write algorithmic steps for Quasi-Newton method. [5]

B. Define: Dual Price [2]
